Design of High Speed Carry Save Adder using Carry Lookahead Adder

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***Abstract*— Addition is one of the essential operations in Digital Signal Processing (DSP) applications which includes Fast Fourier Transform (FFT), Digital filters, multipliers etc. With the advancements in technology, research is still going on to design a adder that performs addition in flash of time. One of such high speed adder is Carry Save Adder (CSA). In this paper we have designed High Speed Carry Save Adder (CSA) using Carry Look ahead adder in the final stage instead of using conventional ripple carry adder [1] so that speed increases by 27.5%.**

Keywords*— *Carry Save Adder, CSA, CLA, RCA, High speed, CSkA, CSelA.

I. INTRODUCTION

Carry save adder is the high speed multi-operand adder. A carry save adder consists of a ladder of full adders. The 8-bit CSA consists of 8 disjoint full adders where each of which computes a single sum and carry bit based on the corresponding bits of the four input numbers. It takes three or more than three 8-bit integers to be added and produces two outputs, 8-bit partial sum and 8-bit carry. Unlike the normal adders such as ripple carry adder a CSA consists of multiple one bit full adders without carry chaining.

Carry Save Adder (CSA) eliminates the carry chaining by saving the carry and provides the carry to the next stage of addition instead of adding it to the next higher bit adder. Here in this paper the CSA is implemented using Carry Look ahead Adder in the final stage instead of using conventional Ripple Carry Adder (RCA).

CSA is simulated on both Xilinx ISE Simulator on Virtex 4 target board and Cadence, and the synthesis results are obtained on Cadence Encounter 180nm technology.

II. WORKING OF CARRY SAVE ADDER (CSA)

1) BASIC WORKING OF CSA

CSA is used to add multiple operands. As considered in the below example three operands are added. The sum & carry generated is similar to Full Adder, but the carry is added to the sum at the next level. At the last stage a ripple carry adder is used to generate sum & carry.

X:		1 0 1 1
Y:	+	1 0 0 1
Z:	+	0 1 1 1
S:		0 1 0 1
C:		1 0 1 1
Sum:		1 1 0 1 1

In the above example three operands x,y,z of four bits are added so that the sum (partial) is calculated and the carry is given to next higher bit and added to the sum (partial) in the next stage. Sum is available after two stages of successive addititions.

2) CONVENTIONAL CSA USING RIPPLE CARRY ADDER

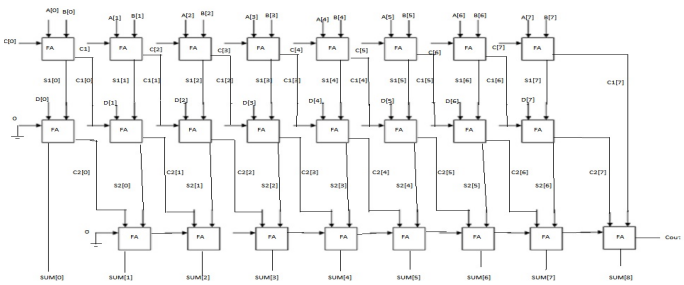

Fig1: CSA using Ripple Carry Adder in final stage

As shown in the above Fig1 CSA is implemented using ripple carry adder in the final stage, which is conventional carry save adder. Here we are adding four operands each of eight bits. In the first stage operands A, B, C are added and we get partial sum1 and partial carryout1. In the second stage operand D is added to partial sum1 we get partial sum2 & partial carry2 & in the final stage we use ripple carry adder to calculate sum and carryout.

III. RELATED WORK

Carry save addition is being used in multipliers, signal processing, FFT etc. But various authors have tried very little to increase the performance in terms of speed.

The conventional CSA [1] as discussed above is used in most of application like booth multiplier, Wallace tree multiplier, parallel multiplier etc. In paper [1] and [2] authors are trying to compare speed or delay, area and power of various adders available like RCA, CLA, CSA, CSkA, CSelA. Out of all the adders author is concluding CSA is best one in terms of speed.

In paper [3] 32-bit array multiplier is implemented using conventional CSA, they are achieving optimized area & also low power but the performance in terms of speed is somewhat less because of ripple carry adder in the final stage.

In paper [5] modified carry save adder for booth multiplier, authors are breaking the final stage of CSA into five groups depending on the number of bits of sum and carry from the previous stages. They are achieving good speed but it involves complexity in designing when compared to our proposed design.

IV. PROPOSED DESIGN

Here in this design normal carry save addition concept is used for first stages and at the final stage carry look ahead adder is used. Use of carry look ahead adder in the final stage increases the speed of addition operation. For each full adder (FA) there are two output bits sum & carry. This FA can be implemented using XOR and AND gates as shown below.

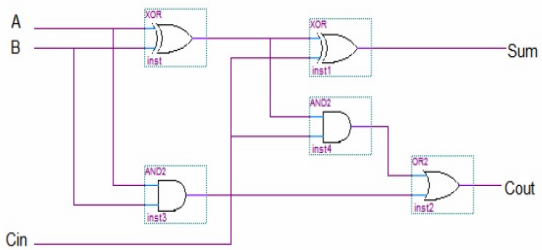

Fig2: Gate level design of Full Adder

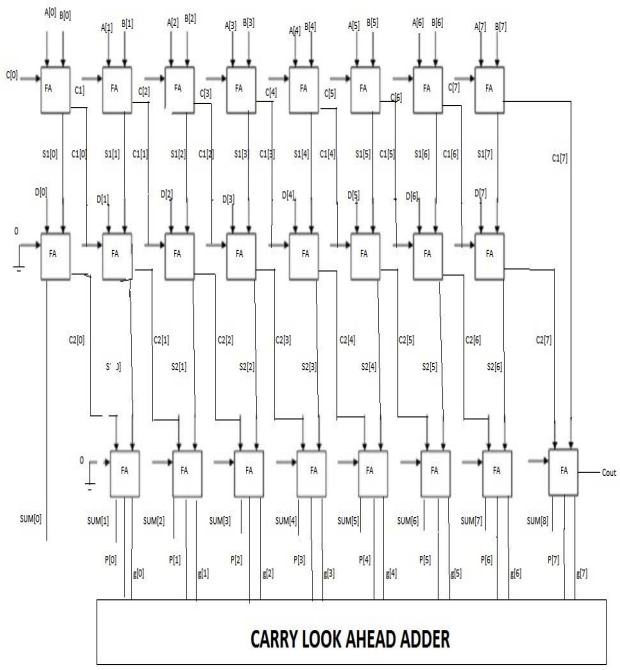

Fig3: Proposed CSA using Carry Look ahead Adder in the final stage

V. IMPLEMENTATION ALGORITHM

1) *Full Adder:*

$$Sum = A \oplus B \oplus C \qquad (1)$$

$$Cout = AB + BC + CA \qquad (2)$$

2) *Carry Look ahead Adder:*

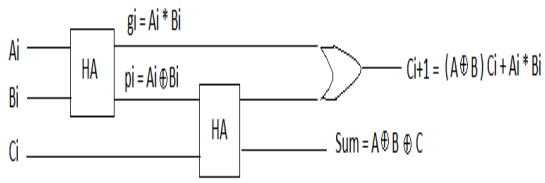

Fig4: Full Adder using two half adders

$$pi = Ai + Bi \qquad (3)$$

$$gi = Ai * Bi \qquad (4)$$

$$Sum = pi \oplus Ci \qquad (5)$$

$$Cout = gi + pi * Ci \qquad (6)$$

We are implementing algorithm or a proposed technique for carry save adder using Verilog HDL. A Verilog code is written for CSA and it is simulated & synthesized further.

VI. RESULTS

1) *SIMULATION RESULTS*

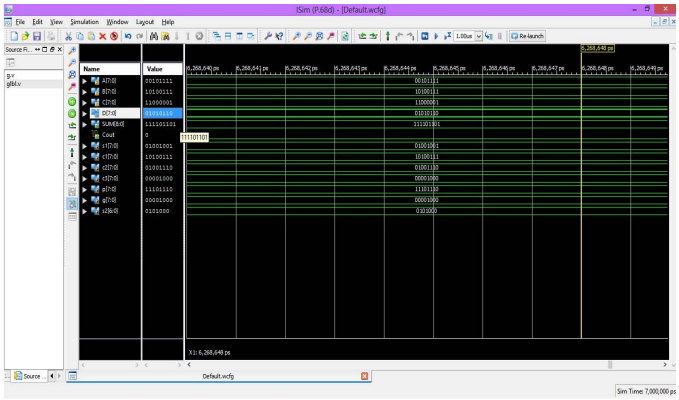

Fig5: CSA using Ripple Carry Adder (RCA) in Xilinx ISE

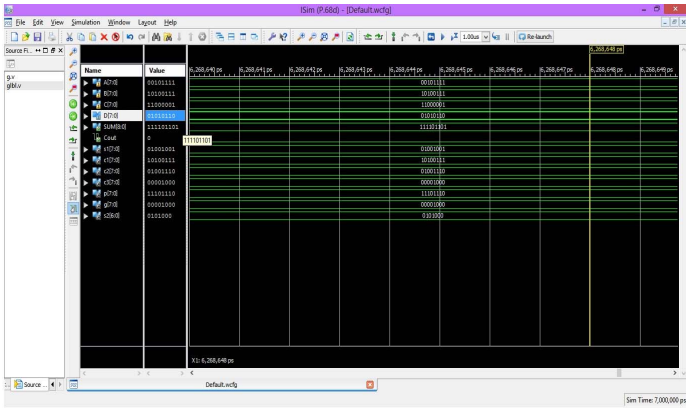

Fig6: CSA using carry Look ahead adder (CLA) simulation in Xilinx ISE

2) *RTL SYNTHESIS RESULTS*

i. *Before RTL synthesis:*

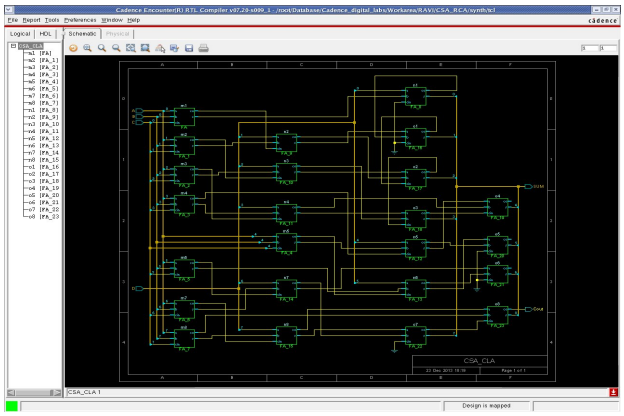

Fig7: CSA using RCA before RTL synthesis in Cadence tool

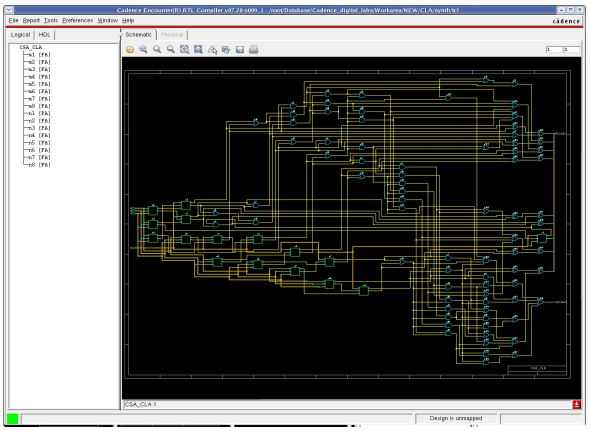

Fig8: CSA using CLA before RTL synthesis in Cadence tool

ii. *After RTL synthesis:*

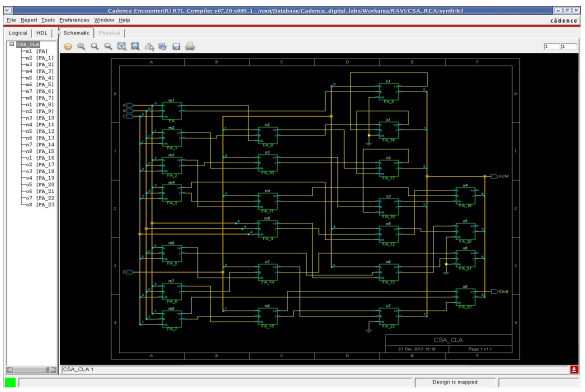

Fig9: CSA using RCA after RTL synthesis in Cadence tool

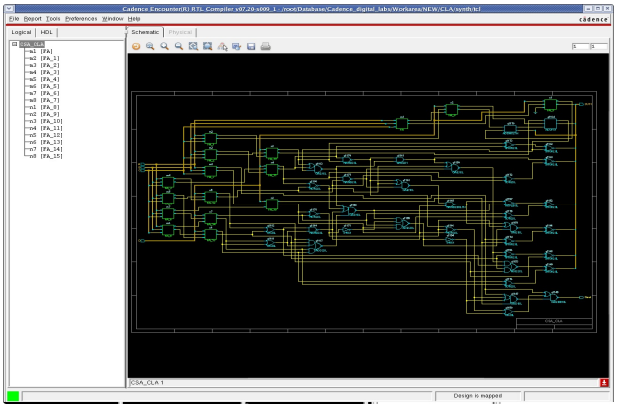

Fig10: CSA using CLA after RTL synthesis in Cadence tool

Simulation results of CSA using ripple carry adder and CSA using carry look ahead adder is shown in Fig.5 and Fig.6 respectively. The functional behaviors of both the designs are same but when we look at the area, power and timing there is a macro difference. As we can see in Fig.9 the area occupied

by CSA using ripple carry adder is less when compared to Fig.10 i.e. CSA using carry look ahead adder. This is because in CSA using carry look ahead adder we are using additional circuitry to increase the speed of operation hence the additional circuitry also consumes additional power which is obvious in VLSI circuits. Always there will be a tradeoff between area, power & timing.

VII. COMPARISON TABLE

Table1: Comparison table between CSA using ripple carry adder & carry look ahead adder

Instance	Cells	Cells Area	Total power(nW)	Timing(ps)
CSA_CLA	54	454	22059.766	1351
CSA_RCA	40	424	20146.463	1865

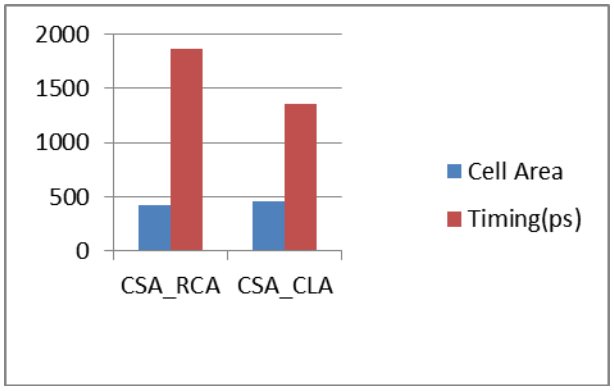

Fig11: Bar graph showing cell area & timing for CSA using ripple carry adder & carry look ahead adder

VIII. CONCLUSION

After comparing the timing results between the carry save adder using ripple carry adder and carry save adder using carry look ahead adder we can conclude that the proposed design i.e. Carry save adder using carry look ahead adder is about **27.5% faster**. So when this CSA when it is used in Digital Signal Processing (DSP), Fast Fourier Transform (FFT) and other applications where timing constraints are crucial where we will be able to meet timing requirements and performance is improved, of course at the cost of some area and power.

IX. FUTURE SCOPE

The proposed design further can be modified by replacing each stage of full adder with carry look ahead adder so that the speed can be increased nearly up to 60% but all this is possible at the cost of some tradeoff between area & power.

If all the three stages of carry save adder are implemented using carry look ahead adder then huge computations involving addition can be solved very fast.

REFERENCES

[1] Maroju SaiKumar, Dr.P. Samundiswary " Design and Performance Analysis of Various Adders using Verilog", in International Journal of Computer Science and Mobile Computing, IJCSMC, Vol.2, Issue.9, September 2013, pg.128-138.

[2] R.UMA, Vidya Vijayan, M. Mohanapriya, Sharon Paul " Area, Delay and Power Consumption of Adder Topologies", in International Journal of VLSI design & Communication Systems, Vol.3, No.1, February 2012.

[3] Raminder Preet Pal Singh, Parveen Kumar, Balwinder Singh, "Performance Analysis of 32 bit Array Multiplier with Carry Save Adder and with Carry Look ahead adder", in International Journal of Recent Trends in Engineering, Vol 2, No. 6, November 2009

[4] Phil Gossett "Quantum Carry – Save Arithmetic", research paper, Silicon Graphics, Inc. 2011 N.Shoreline Blvd. Mountain View, Ca 94043-1389, August 29,1998.

[5] S.Murugeswari, S.Kaja Mohideen "An area efficient and low power multiplier using modified carry save adder for parallel multiplier", in Springer-VerLAG Berlin Heidelberg, V.V Das and Y.Chaba (Eds): AIM/CCPE 2012, CCIS 296, pp.456-460, 2013

[6] James Levy and Jabulani Nyathi " AHigh Performace, Low Area Overhead Carry Lookahead Adder", research paper, Washington State University, School of Electrical Engineering and Computer Science, Pulaman,WA 99164

[7] Magnus Karlsson, " A Generalized Carry Save Adder Array for Digital Signal Processing", research paper, OKG AB, SE – 57283.

[8] Taewhan Kim, William Jao, Steve Tjiang "Arithmetic Optomization using Carry Save Adders" in 35th Design Automation Conference, DAC98- 06/98 San Franscisco, CA USA

[9] Neil Weste and K Eshragian, "Principles of CMOS VLSI Design: A system Perspective, "2nd edition, Pearson Education(Asia) Pvt. Ltd, 2000.